SECTION 15934
ROOFTOP AIR HANDLING UNITS FOR CORRIDOR
(CONSTANT VOLUME SYSTEM)

PART 1 – GENERAL

1.01 DESCRIPTION OF WORK

A. Provide packaged constant volume air handling units with four pipe hydronic 30% propylene glycol heating and cooling coils as specified, as shown on the Drawing schedules and as needed for a complete and proper installation. Product specific requirements are contained herein; Section 15501, General Provisions for Heating, Ventilating and Air Conditioning Work, shall be referred to for general requirements.

B. Constant volume air handling units shall be provided for the corridor as shown on the Drawings.

C. All fans of constant volume air handling units shall be provided with variable frequency drives to facilitate the initial balancing procedure and for fan airflow control per the Sequence of Operation.

D. Air handling units shall be provided with OEM controls; OEM controls shall be integrated into the BACnet network by the Temperature Controls Contractor (TCC).

1.02 COMMISSIONING REQUIREMENTS

A. HVAC Contractor shall comply with the Commissioning Requirements of Contract Specification Section S01660 and Section {15971}.

1.03 INDOOR AIR QUALITY (IAQ) REQUIREMENTS DURING CONSTRUCTION

A. During construction, the HVAC Contractor shall comply with the Indoor Air Quality (IAQ) Requirements indicated in Contract Specification Section S01550.

1.04 RELATED SECTIONS

A. Division 15 Sections

B. Division 16 Sections

1.05 SUPPLEMENTAL SUBMITTALS

A. Product Data

1. Manufacturers Product Data (including motor manufacturer affidavit if greater than 60Hz operation)

2. Installation Instructions

3. Performance Charts

B. Shop Drawings

1. Submit assembly-type shop drawings showing unit dimensions, weight loadings, required clearances, location, construction details, and mounting and field connection details.

2. Wiring Diagrams: Submit manufacturer's electrical requirements for power supply wiring to air handling units by Division 16 Electrical Contractor. Submit manufacturers wiring diagrams for interlock and control wiring. Clearly differentiate between portions of wiring that are factory-installed and portions to be field-installed by the Temperature Control System Contractor (TCC). Submit BACnet control interface and control wiring with Sequence of Operation.

C. Coordination Drawings: Submit with Shop Drawings. Show layout and relationships between components and adjacent structural and mechanical elements. Show support locations, type of support, and weight on each support. Indicate and certify field measurements.

D. Certifications

1. Include the certified fan-sound power ratings. Provide radiated, supply discharge and return inlet computer generated sound power levels.

2. Contractor’s start-up and demonstration affidavit.

3. Certification of operability for smoke control systems that are to remain operable following the

design earthquake ground motion per Section 13.2.2 of ASCE 7-**16.**

E. Maintenance Data

1. Maintenance Manual

F. Maintenance Materials: Filters and belts (when belt driven fans are provided in lieu of direct driven fans).

G. Preventative Maintenance Plan and Schedule

H. Warranties specified in this Section.

I. All air handling unit end devices including but not limited to VFD drives for total enthalpy wheels and heat wheels, outside air intake dampers, exhaust dampers, recirculation dampers and heating/cooling valves (shipped loose if necessary) shall be factory tested by simulating their hard-wired sensor input and observing/recording the proper operation of the end device. Supply fan and exhaust fan VFDs shall be factory configured/programmed and tested. Submit all factory testing results.

J. The Original Equipment Manufacturer (OEM) shall furnish to the Authority a licensed fully-functional copy of the toolset (i.e. software program) necessary for communication with and the adjustment of their controller.

1.06 SUPPLEMENTAL QUALITY ASSURANCE

A. Codes and Standards

1. All appliances regulated by the New York City Construction Codes shall be listed and labeled (reference Section MC **301.7** of the **2022** NYC Mechanical Code). Testing of material shall be in accordance with Section §28-**113.4.1** of the **2022** NYC Administrative Code (reference Section MC **301.8**). Whenever the NYC Construction Codes or the Rules of the Department of Buildings requires that material be listed or labeled and material proposed to be used is not so listed or labeled, the use of such material shall be subject to prior approval by the Commissioner (Office of Technical Certification and

Research OTCR) and such material shall be used only to the extent set forth in such approval. Materials that were previously approved by the Board of Standards and Appeal (BSA) or by the Department (MEA) before the effective date of the NYC Construction Codes may continue to be used, but only to the extent set forth in such approval, and only if such approval is not specifically amended or repealed by the Commissioner.

2. Packaged air handling units shall comply with the 2020 New York City Energy Conservation Code (NYCECC) and ASHRAE 90.1-2016 as modified by Appendix CA of the 2020 NYCECC.

3. A prototype unit shall be ETL listed and labeled, classified in accordance with UL 1995-**2011**/CAN/CSA/No. 236-M90.

B. Before submitting any equipment shop drawings for approval, the HVAC Contractor, Temperature Controls Contractor (TCC) and the Equipment Vendor and Manufacturer shall coordinate the controls required for the system.

C. Wiring internal to the unit shall be wired to a numbered terminal strip for simplified identification and ease of trouble shooting.

D. The manufacturer shall have a quality management system in place equal to the quality assurance standard ISO-9001, for the design, manufacture, and service of heat wheels, energy recovery wheels and packaged ventilation/air conditioning equipment. Units that are field assembled from pre-fabricated panels shall not be considered or accepted. <u>The OEM shall have been a manufacturer of packaged energy recovery equipment for at least ten years prior to bid time.</u>

E. Fan performance shall be based on tests and procedures performed in accordance with AMCA Publication 211-**2022** and Publication 311-**2016** and comply with the requirements of the AMCA Certified Ratings Program. Fans shall bear the AMCA seal.

F. All wheels shall have prototypical performance testing in accordance with ASHRAE Standard 84-**2020** and **AHRI**

Standard 1060-**2018.** The prototype testing shall be conducted by a qualified independent testing laboratory. Prototype testing shall confirm published performance and document that the wheel material does not transfer pollutants typically encountered in the indoor air environment. Provide the prototype testing reports.

G. Casing paint system shall comply with ASTM B117 2000-hour salt fog resistance test.

H. Fan motors shall be furnished with efficiencies equal to or greater than those specified in the Energy Policy Act of 1992 (EPACT), the 2020 NYCECC and ASHRAE 90.1-2016 as modified by Appendix CA of the 2020 NYCECC.

I. All hot water and chilled water coils shall be rated in accordance with **AHRI** standards.

J. Required smoke dampers for 15,000 CFM and greater capacity shall be classified by Underwriters Laboratory as a leakage rated damper for use in smoke control systems under UL555S**-1999 or later version** and shall be minimum leakage Class II, rated for minimum 250°F per Section MC **607.3.2.2.**

K. Filters shall comply with UL 900-**2004** per Section MC 605.2.

1.07 WARRANTY

A. In addition to the warranties specified in Section G01740, the Contractor shall provide a written warranty signed by manufacturer, agreeing to replace/repair components that have inadequate and defective materials and workmanship including leakage, breakage, improper assembly, or failure to perform as required, within the warranty period.

B. The warranty shall cover all parts and labor for a period of two years from the date of Substantial Completion. Exceptions are as follows: All VFDs shall be warranted for parts and labor by the manufacturer for a period of 36 months. All total enthalpy energy recovery wheels and heat wheels shall be warranted for parts and labor by the manufacturer for a period of 60 months. Provide a parts and labor warranty for the casing of the unit for up to 25 years against complete rust through and/or

surface rust that exceeds 60% of any given panel. The manufacturer can reserve the right to deny any casing rust claims if there have been any modifications to the original cabinetry (i.e. removing graffiti, cleaning practices other than recommended, any dents acquired in the field, etc.). All warranties shall use Substantial Completion as the warranty start date. The warranty shall include parts, labor, and travel costs incurred by the manufacturer to provide factory authorized on-site services. The Contractor shall be responsible for emergency repair during the first 24 months of the system operation. Contractor shall use factory authorized repair personnel. Equipment submittals will not be approved until submission of the manufacturer's warranty indicating compliance with the above.

1.08 MAINTENANCE

A. Provide with each packaged air handling unit, one spare set of air filters. Suitably box and label spare filters as to their usage. Also provide spare belts for belt-driven equipment.

1.09 COORDINATION

A. Coordinate size, location and installation of rooftop air handling units and dunnage supports with roof installer.

1. Coordinate installation of vibration isolation and seismic restraints (Reference Section 15503 or 15504) for all air handling units.

B. It is the intent of this Article to clarify that the Temperature Controls Contractor (TCC) is responsible for the incorporation of all controllers into the BACnet MS/TP or BACnet IP network for controllers furnished both by the OEM and by other vendors/manufacturers. It is recognized that controls furnished by other vendors/manufacturers and by trades other than mechanical necessitate that both the General Contractor (GC) and Mechanical Contractor (MC) actively facilitate the coordination of controls between all parties. It is the responsibility of each vendor/manufacturer furnishing a digital controller for its proper operation, configuration/programming, and compliance with all appropriate specification sections and for the

complete documentation of the vendor's/manufacturer's communication cards. It is the responsibility of each vendor/manufacturer to comply with the intent of this specification section relating to, but not limited to providing direct, technical assistance to the TCC to integrate the vendor's/manufacturer's device (s) into the network, technical performance, warranties, service responses, etc.

C. Provide control devices as coordinated by the Mechanical Contractor (MC) and the Temperature Control Contractor (TCC) to achieve required seamless integrated controls.

D. Air handling units shall be provided with Original Equipment Manufacturer (OEM) furnished and installed controls. OEM controls shall be native BACnet or Utilize a non-Native BACnet Controller with a BACnet Communication Card to be integrated into a BACnet network.

E. The I/O points to be furnished are as described on the Drawings. Specification Section 15985 describes the required Sequence of Operation. The unit shall be delivered to the site ready for connection of field-based control devices and network connections.

1.10 PREVENTATIVE MAINTENANCE

A. Provide a two (2) year full preventative maintenance contract on the air handling unit (AHU) equipment. The two-year term shall commence at Substantial Completion of the project or after submission of correctly completed applicable guarantee and warrantee documentation for the units, whichever is later. The Contractor shall perform all maintenance on the equipment until start of the preventative maintenance contract.

1. The firm engaged for the preventative maintenance contract must be authorized by the equipment manufacturer to provide such maintenance. The maintenance work shall be performed by trained personnel directly employed and supervised by manufacturer's authorized maintenance contractor.

2. The cost of this preventative maintenance contract shall be included in the bid price.

3. This maintenance contract shall be managed and administered by the Department of Education (DOE)

4. The preventative maintenance requirements for the specified equipment shall be in accordance with the equipment manufacturer's preventative maintenance requirements (as defined in the manufacturer's Installation and Operations Manual) and as specified herein.

5. Perform scheduled maintenance work during regular working hours. All work shall be coordinated with the Custodians Office.

B. Preventative Maintenance Procedures:

1. Preventative maintenance procedures shall follow all safety and O.S.H.A. requirements.

2. The Contractor shall submit for the Designer of Record's approval a preventative maintenance plan and schedule, conforming to either equipment manufacturer's preventative maintenance requirements and schedule or as indicated below, whichever is more rigorous and stringent. All system components requiring preventative maintenance shall be included. The plan shall identify:

   a. the equipment and manufacturer

   b. components requiring maintenance

   c. type and description of maintenance to be performed.

   d. maintenance interval

   e. maintenance contractor approved by the equipment manufacturer (include certifying documentation by the equipment manufacturer)

   f. component replacement schedule as applicable (i.e. belts, bearings etc.)

   g. filter replacement schedule

h. lubrication schedule

i. lubrication types

3. The plan shall include a copy of the equipment manufacturer's published maintenance requirements and schedule.

4. The costs of all parts requiring periodic replacement (i.e. filters, belts, etc.) are to be included in the preventative maintenance contract. Supply all necessary lubricants, filters, cleaning materials, etc. as required to keep the system in good working order during the preventative maintenance period.

5. Maintenance of equipment and parts shall be either as per the manufacturer's maintenance requirements and schedule as published in their Installation and Operations Manual or as indicated within herein, whichever is more rigorous and stringent. Any preventative maintenance defined in the manufacturer's Installation and Operations Manual for components not defined below shall be followed and included in the plan. The interval between preventative maintenance visits shall be as defined in the manufacturer's Installation and Operations Manual, but in no case shall the interval between maintenance visits be greater than three (3) months.

a. AHU - Casing/Cabinet Integrity

1) Inspect for defective integral dampers, access doors, clogged louvers and bird screens.

2) Clean, align and lubricate as required.

3) Replace screws, linkage or latches and patch or replace gaskets.

4) Inspect for broken, loose or missing connections and parts. Correct as required.

5) Inspect for improper setting of outside air, recirculation and exhaust dampers. Correct as required by cleaning and lubricating pivot points and actuator mechanisms.

6) Inspect guards, casing, supports, platforms, and mounting bolts.

7) Inspect for water leaks through air handling unit enclosure. Seal as required.

b. AHU – Filters

1) Replace pre-filters and final filters as per manufacturer's maintenance schedule, but as a minimum at every 6 months.

2) Check and log pressure drop across the outside air filter set and return/exhaust filter.

c. AHU – Supply and Exhaust (or Return) Fans & Motors

1) Perform maintenance at each visit. Report all findings and non-maintenance deficiencies to the custodian and note in the log.

2) Remove all debris from fan section and unit in general and check all fan safeties including but not limited to thermal overloads and static pressure sensors. Adjust as needed.

3) Visually inspect all fans. Make sure nothing is blocking the fan blades and check for obvious signs of damage.

4) Measure each fan's flow by measuring the total static pressure change across the fan and measure fan shaft rotation with a tachometer. Use manufacturer's fan curve to determine the flow based on

measured fan speed and pressure change across the fan.

5) Verify that the fan motor amperage is compliant with manufacturer's fan curves for the measured flow magnitude. Verify that the fan motor is running in the correct direction.

6) Check that the wire leads on the motor have not been switched.

7) Check for excessive bearing temperatures, noise and vibration indicating inadequate lubrication of bearings and moving parts.

8) Fill all grease dispensers or oil cups at each visit. If fan bearings are sleeve bearings, lightly oil with the recommended lubricant (Do not over lubricate). Follow manufacturer's lubrication interval and recommended lubricant type. Fans equipped with self-lubricating bearings (sealed-cassette ball-bearing cartridges preloaded with grease) are not to be re-greased.

9) Grease any conventional ball bearings. Open the drain plug and inject grease through the fill fitting until clean grease comes out of the drain.

10) Clean fan/impeller blades.

11) Adjust belts to provide proper belt tension. Belts should be aligned with a straightedge to prevent lateral wear. Change belts as required depending on wear. All belts shall be replaced as a minimum at every year upon commencement of the maintenance contract.

12) Check fan vibration isolators.

d. AHU – Chilled Water Coil

1) Clean chilled water coil at each visit. Clean/brush with a quality cleaning solution allowing an appropriate dwell time necessary to effectively remove all microorganisms/ biofilms. Steam clean coil if required to remove any residual build-up. The entire air flow path shall be treated with a germicide once it is cleaned to assure the unit remains free of microorganisms between cleanings.

2) Check condensate drain piping (multiple connections) and verify condensate drain pans are clean. Log each coil air temperature change and each coil air pressure drop.

e. AHU – Hot Water Coil

1) Clean hot water coil at each visit. Brush and/or use compressed air or steam clean or vacuum as required.

2) Report all findings and non-maintenance deficiencies to the custodian and include in the maintenance log.

f. AHU – Integral Smoke Detector/Smoke Damper/Fire Damper

1) Perform a minimum of twice a year.

2) Check and make sure smoke detectors and dampers are functioning properly in the AHU. Check smoke detectors with magnet or smoke can. Check dampers close upon activation from the Fire Alarm Control Panel. Report all findings and non-maintenance deficiencies to the custodian and include in the maintenance log.

g. AHU – Piping

1) Inspect for corrosion at each visit and report all findings and non-maintenance deficiencies to the custodian and include in the maintenance log.

h. AHU – Wiring/Power

1) Inspect for wiring and electrical defects at each visit. Inspect for loose connections; charred, frayed, broken, or wet insulation; short circuits; lose or weak contact springs; worn or pitted contacts; and other deficiencies.

2) Adjust settings; tighten connections; clean contacts and adjust alignment.

3) Inspect and test all power supplies. Record all voltage readings. Check all relays and terminal connections. Clean/wipe down panel face. Check variable frequency drives on units.

4) Variable Frequency Drives (VFD): Check integral cooling fan for functionality. Clean air inlet filter. Vacuum off dust from printed circuit boards.

5) Report all findings and non-maintenance deficiencies to the custodian and include in the maintenance log.

i. AHU – Total Enthalpy Energy Recovery Wheel and Heat Wheel

1) Perform maintenance at each visit.

2) Clean the media with low pressure steam (less than 5 psi) or compressed air (20 psig maximum) or hot water (provide drain pan to collect the water) or light detergent.

3) For non-sealed bearings, the entire rotor and wheel assembly shall have the main bearings greased biannually.

4) Adjust belts to provide proper belt tension. Belts should be aligned with a straightedge to prevent lateral wear. Change belts as required depending on wear. All belts shall be replaced as a minimum at every year upon commencement of the maintenance contract.

j. AHU - Integral OEM Provided Sensors

Check sensor calibration at each maintenance visit and recalibrate if necessary.

k. AHU - Integral control valves

1) Perform a minimum of twice a year.

2) Operate integral valves from full open to full closed to assure operability.

3) Check flange bolting and actuator mounts for evidence of loosening and correct as needed.

4) Inspect the valve and surrounding area for previous or existing leakage at flange faces or shaft connections.

5) Check wiring to actuators and related equipment for looseness and correct as needed.

6) Make service recommendations to address any binding valve stems and leakage. Recommendations might include disassembly/cleaning of valve internals, polishing the binding stem by removing scratches and mineral buildup and replacing the valve seat discs. Report all findings and non-maintenance deficiencies to the custodian and include in the maintenance log.

6. During each preventative maintenance visit and in addition to the preventative maintenance recommended by the equipment manufacturer, the qualified technician shall also activate and inspect all components of the equipment for wear, leaks, condition, operation, and report findings in the log and directly to the custodian.

7. During each preventative maintenance visit, the maintenance technician shall make all necessary adjustments as necessary to meet the performance requirements as specified in the manufacturer's Installation and Operation Manual.

C. The preventative maintenance contractor shall advise the custodian of every preventative maintenance visit, including any material that was used, as well as any deficiencies noted in the system.

1. The technician shall maintain a visit log(s) indicating, for each unit covered under the maintenance contract, the following:

a. date of visit

b. the unit (name/location etc)

c. component(s) receiving the maintenance

d. maintenance performed

e. lubrication performed

f. parts and filters replaced

g. condition and operability of all components

h. date of next required maintenance.

i. deficiencies noted and reported to the custodian.

2. Copies of log(s) shall be kept in the custodian's office. The originals shall be maintained by the maintenance contractor until the expiration date of the maintenance contract at which time the originals shall be turned over to the custodian.

D. End of the Preventative Maintenance Contract

Approximately three (3) months prior to the end of the maintenance contract term, the DOE and the preventative maintenance contractor shall make a thorough inspection of all air handling unit equipment covered under the Contract. At the conclusion of this inspection, the DOE will give the preventative maintenance contractor written notice of any maintenance deficiencies for which the maintenance contractor was responsible. The preventative maintenance contractor shall be responsible for correction of these maintenance deficiencies within 30 days after receipt of such notice. Other than the referenced walk through prior to the end of preventative maintenance period, the DOE may conduct inspections during the 2-year preventative maintenance period and the preventative maintenance Contractor shall comply and clear any maintenance deficiencies found within 30 days after receipt of such notice.

PART 2 – PRODUCTS

2.01 PACKAGED AIR HANDLING UNITS

A. General

Provide factory assembled and tested packaged air handling units as indicated on the Drawings. Units shall be especially designed, engineered and manufactured for outdoor installation.

1. In the event the packaged air handling unit is shipped and assembled in multiple pieces, the packaged air handling unit manufacturer shall provide the services of a factory engineer and technical support to supervise the re-wire, re-test, and re-assembly of the packaged air handling unit in the field. Prior acceptance of the SCA is required for packaged air handling units that cannot be manufactured, shipped, and rigged as a complete assembly.

2. Factory operational test sheets shall be provided to the SCA. The SCA reserves the right to witness all factory testing and shall be given two weeks notice before any factory testing is conducted by

the packaged air handling unit manufacturer. The factory testing and inspection is to be included in the schedule and the manufacturer/Contractor is to accommodate.

3. A packaged prototype air handling unit shall have been tested and ETL listed as a complete packaged unit and therefore all units shall bear the ETL label. All parts shall be available at the local wholesaler level.

B. Capacities/Characteristics

Provide capacities, electrical characteristics and component modules as indicated on the Drawing schedule of equipment.

C. Variable Frequency Drive (VFD)

(VFD BACnet Card and VFD shall be provided by the Air Handling Unit Manufacturer).

1. Packaged air handling unit supply and exhaust fan motors shall be provided with solid state variable frequency drives to convert three phase, 60 hertz power to adjustable frequency and voltage three phase power for stepless motor speed control from 0% to 100% of base speed. Input voltage shall be 460-3-60 or 208-3-60 as indicated on the Contract Drawings. The VFD shall be capable of operating within the range of 10% over to 10% under the rated input voltage. The VFD shall be designed for use on variable torque loads and shall be capable of providing sufficient torque to start motor from rest with initial influx of power. The VFD operation shall be fully factory tested prior to shipment and installation. VFDs shall be used for fan balancing and fan airflow control per the sequence of operation. The VFD controller shall have the following features: isolated power for control circuits, automatic type reset faults for over-current, under-voltage, over-voltage, phase loss or over-temperature, fault counter reset, adjustable current limiting circuitry, independent acceleration and deceleration time adjustment, 4-20 mA control signal interface, 0-10 Vdc output signal, and automatic frequency adjustment. VFD shall be

provided with digital read output meter indicating percent speed and load, local/remote selector switch, power/on, diagnostic, and fault trip indicating lights, Hand/Off/Auto selector switch, manual by-pass switch, by-pass starter, and manual speed control adjustment. The complete drive shall be tested to ANSI C37.90.1 and UL Standard 508 and be ETL or UL listed. Power line noise distortion shall be limited to voltage distortion factor and line notch depth defined in IEEE Standard 519-1981. Total voltage distortion shall not exceed 5%. VFD shall be provided with two normally open and two normally closed dry contacts rated at 120 volts, 10 amps, 60 Hz for remote indication of the following: controller shutdown with automatic restart, system shutdown with automatic restart, and system running condition. Controller shall include a door interlock fused safety disconnect or circuit breaker that will disconnect all input power. VFD shall be equipped with a line reactor electrical noise suppression filter to limit the electrical noise on the data communication channels.

The VFD supply and exhaust fan packages shall be selected with inverted rated duty motors and motor winding insulation complying with IEC Part 34.17.

2. Variable Frequency Drive (VFD) BACnet Card

The VFD drives for variable speed supply and exhaust fans shall be provided with BACnet interface cards to provide the BACnet open protocol for control communications with "plug & play" capabilities. The BACnet interface card shall use BACnet objects to communicate on a BACnet network, as defined in the BACnet functional profile for Variable Speed Motor Drives.

The BACnet interface card shall be designed and constructed to comply with the BACnet Interoperability Guidelines and the BACnet Functional Profile for a 3-phase asynchronous motor drive.

The BACnet interface card shall be mounted integral to the drive and shall provide an easy connection point for the BACnet network media. The BACnet

interface card shall be factory installed by the VFD unit manufacturer.

The BACnet interface card shall be compatible with BACnet twisted pair Master Slave Token Passing (MS/TP) or BACnet IP. The VFD unit manufacturer shall provide a polarity sensitive, pluggable 3-way Phoenix- style connector or a modular RJ-45 plug, to provide primary network connection of the VFD drive to the BACnet control system. The option card receives its power from the drive control board.

The BACnet interface card shall be provided with diagnostic lights to assist in the configuration and commissioning the VFD onto the BACnet network.

Provide network wiring between the VFD BACnet interface cards and any BACnet unit controller(s). Connect the VFD BACnet interface card to the network. Terminate the BACnet network wiring to marked terminal strip for use by the TCC in making the field network connections.

The VFD BACnet interface card shall provide the following network features:

a. Access to all drive parameters

b. Access to all drive commands and diagnostics

c. Monitor/Command Points –Provide List for the Commissioning Authority’s Approval. See Note 1 below.

d. PID Feedback Through Network

e. Selectable Run/Reference Command

f. Drive Ready Status Indication

g. KWH, KW Meter

h. Run Time Meter

i. Custom Display

j. Programmable Cable Loss Detection (Fault, Last Speed, Preset Speed)

k. Safety Interlock Detection (Bypass)

The VFD unit manufacturer shall provide the following VFD drive system characteristics for the Commissioning Authority's approval:

1) List of the VFD Object Network Variable Summary

2) List of the VFD Object Configuration Properties

3) List of the VFD Node Object Operations and Object Requests/Status

4) List of the VFD Standard BACnet Object Types

5) List of the VFD Drive network variables for monitoring and operating status

3. Hand/Off/Auto functionality shall be built into the VFD digital operator keypad for operation from remote (Auto) or local (Hand) signals. A copy function shall be provided for set-up parameters when programming multiple drives. Factory "default" parameters shall be user selectable for typical fan applications and shall be able to be field refined for custom applications.

The VFD shall be provided with a PI controller that shall maintain setpoint for control of the fans based on pressure, flow, temperature or manual regulation.

a. Network Installation: The VFD unit manufacturer shall include with each VFD drive an integral BACnet interface card and a diskette containing the required device files to enable node installation and identification onto the BACnet network. The VFD unit manufacturer shall provide the External Interface File (XIF) and the Device Resource

Files (TYP) for field network connection by the Temperature Control Contractor.

b. External Interface File: The VFD External Interface File shall contain a detailed description of the VFD node's network variables and configuration properties, including network variable type and self-documentation information. Hardware characteristics, such as transceiver type and initial communication parameters, shall also be defined.

c. Device Resource Files: The Device Resource File shall define the components of an external interface for the VFD. The files shall provide installation tools and operation interface applications to interpret and format the VFD data properly. The VFD Standard Resource Files shall contain the definition for Standard BACnet Object types, Standard Configuration Property Types and Standard Functional Profile Types compliant with the requirements of the BACnet Interoperability Association. The resource files shall be included on the installation disk, provided by the VFD unit manufacturer, and shall contain information specific to the VFD interface requirements to the BACnet object types.

These files shall be placed in the TYPES sub-directory associated with the VFD device network management tool and registered in the VFD resource file catalog.

4. Acceptable VFD Manufacturers

Yaskawa
Asea Brown Boveri (ABB)
Siemens
Danfoss
Vacon
GE Fuji
Square D
Or approved equal

D. Packaged Air Handling Unit Construction

1. Casing

a. Base Frame: The base of the package shall be an all-welded structural "C" channel steel frame or rectangular steel tubing with required tubular and angular cross-members as required to maintain floor rigidity and stiffness and act as isolator supports. Base shall be all solid welded. The base shall be painted with one coat of a lead-free, rust-inhibiting, two-part epoxy primer, followed by two coats corrosion and weather resistant urethane modified enamel paint. Lifting lugs shall be an integral part of the structural frame and shall be all welded using three-pass welds. Frame shall be constructed to limit deflection to 1/200th of its span in any direction. Bolted or screwed base frame and floor assemblies are not acceptable due to their poor structural integrity and propensity to leak. All rooftop air handling units shall be mounted on steel dunnage unless otherwise noted.

b. Cabinet: Unit casing shall be of the monocoque stressed skin design (construction technique that supports structural load by using an object's exterior) or casing can optionally be composed of an internal frame or truss that is then covered with a non-load-bearing skin. Casing shall have a minimum 2" thick (when 2 lb/ft^3 density polyurethane foam is used) double-wall watertight construction for all walls, floor (see alternate floor construction below) and roof. Non-load bearing walls and roof shall have an 18-gauge galvanized steel or 16-gauge aluminum outer skin (see casing exterior finish options below) with a 22-gauge galvanized steel or 20-gauge aluminum inner liner. Perforated lining shall be provided in the fan sections for additional sound attenuation. If casing is load bearing, exterior shall be minimum 16-gauge galvanized steel or 14-gauge aluminum and interior shall be minimum 18-gauge galvanized steel or 16-

gauge aluminum. Perforated lining shall be provided in the fan sections for additional sound attenuation. 2", 2 lb/ft^3 density polyurethane foam with an R-value of 6 per inch shall be secured between the inner and outer skins. 1.5 lb/ft^3 density or 3.0 lb/ft^3 density non-eroding rigid board fiberglass insulation may also be used in lieu of the polyurethane foam but the 2" double wall thickness specified above shall be adjusted upwards to achieve the same overall R values as achieved with the polyurethane foam. The insulation shall be held between the inner and outer walls and shall not be exposed to any air streams. All seams shall be positively sealed to prevent water and air leakage. All joints shall incorporate a flanged and capped standing seam to insure a watertight seal. A full perimeter rain gutter shall be provided for outdoor units. A thermal break shall be provided where unit panel sections are joined together to form a wall or floor.

Air leakage shall not exceed 1% at 1½ times maximum unit operating pressure. Air leakage factory testing shall be completed on at least one air handling unit per project and the factory testing results shall be submitted.

All fastening hardware between wall panels shall be encapsulated within the wall for a clean exterior appearance and to minimize exterior wall panel penetration. Unit shall be constructed to limit panel deflection to 1/200th of its span in any direction.

c. Access Doors: Self-supporting hinged access doors shall be provided for all internal components requiring periodic maintenance or inspection. Access panels in lieu of access doors are unacceptable. Weather-resistant closed-cell neoprene bulb type gaskets shall be a mechanically fastened to the frame to create a smooth continuous seal, with long radius corners, around the entire perimeter of the door with only a single splice point. Single piece long radius gasket material shall

be provided to prevent sharp corners and multiple splice points. The door shall be insulated the same as the unit casing, and with the same double-wall construction as the unit casing and shall have full-length stainless steel or aluminum piano-type hinges or stainless steel or aluminum butt hinges for rigidity and airtight enclosure. A minimum of two adjustable door latches shall be furnished for each hinged door. The latches shall have a single handle arrangement that allows the use of one handle to rotate multiple cams. Door handle shall be provided with large nylon tapered roller cam for ease of operation and superior gasket depression. Doorframes shall be a minimum 16-gauge aluminum or aluminized steel or 304L stainless steel, welded at the corners to create a continuous rigid sealing surface for the door. All exterior doors shall be equipped with rain gutters. Doors shall be manufactured in such a way as to allow removal of interior components without disassembly of the wall panels. Hinged doors shall include stainless steel door tie-backs. All access doors shall be provided with provisions for padlock hasps for securing access to authorized personnel only. Access doors shall be provided with 9″ X 9″ double pane Plexiglas windows in the fan sections. All access doors shall open against pressure.

d. Floors: Alternate acceptable construction techniques:

1) Option 1: Provide minimum 2″ double wall floor construction. Walk on floor material shall be a minimum of 12-gage (0.081″) aluminum. Sub-floor shall be 22 gage (0.030″) galvanized G90 steel. 2″, 2 lb/ft^3 density polyurethane foam with an R-value of 6 per inch shall be secured between the inner and outer skins. 1.5 lb/ft^3 density or 3.0 lb/ft^3 density non-eroding rigid board fiberglass insulation may also be used in lieu of the polyurethane foam but the 2″ double wall thickness specified above shall be

adjusted upwards to achieve the same overall R values as achieved with the polyurethane foam. The insulation shall be held between the inner and outer walls and shall not be exposed to any air streams.

2) Option 2: Floors shall be constructed of seam welded 16-gage (0.060”) aluminized steel. Floors shall have an upturned flange around the entire perimeter and around all interior chases to contain moisture within the unit. The underside of the floor shall have a minimum of 1” thick sprayed urethane foam insulation

e. Casing Finish

Option 1: Exterior G90 galvanized or aluminum wall panels shall be pressure wash cleaned and degreased prior to application of primer. Paint system shall include 0.15 minimum average mil primer and 0.75 minimum average mil textured finish top-coat. Wall panels shall be positively coated, including all seating surfaces and the inside surface facing wall insulation (painted before assembly). Paint system shall comply with ASTM B117 2000-hour salt fog resistance test.

Option 2: Exterior walls shall be pre-painted Galvalume Plus mill finish corrosion resistant panels. Paint system shall include 0.15 minimum average mil primer and a 0.75 minimum average mil textured finish top-coat. Manufacturers that do not have proper tooling and experience with forming Galvalume Plus shall provide paint system in compliance with ASTM B117 2000-hour salt fog resistance test.

f. Hoods: For outdoor units, intake and exhaust hoods shall be provided. Opening shall be covered with an aluminum bird screen which is separate from the hood. The intake hood shall have a maximum 500 feet per minute face velocity through the free area. Hood material shall match that of the outer casing, and may

be shipped loose for field installation and/or assembly by the Contractor.

g. Split Unit Construction (if unit must ship in sections - requires approval by Engineer of Record): Each section shall be protected during transit with plastic wrap completely sealing the internal portions of all air tunnels. Each section shall include lugs for rigging. Full height structural steel cross members shall be around the entire perimeter of each section. All welded bar plate shall be located adjacent each split at the base level. Bar plate shall be pre-drilled for acceptance of factory-supplied hardware for re-assembly. The unit floor and roof shall include upturn flange at split location and factory supplied clip shall be positioned over flange for sealing. Factory supplied splice plates shall seal inner and outer walls. All hardware, fasteners, sealant and detailed instructions shall be factory supplied. Internal wiring that traverses each unit split shall include junction boxes in each section. Jumper wire shall be included and neatly bundled in one of the junction boxes. Terminal blocks, or means for wire reconnection shall be factory provided. Equipment manufacturer shall indicate on the equipment submittal where the shipping split is to occur.

2. Total Enthalpy Wheel and Heat Wheel

a. Wheel Matrix: Rotors shall be constructed of rotating honeycomb aluminum matrix. Energy wheel shall have a highly selective desiccant permanently bonded to aluminum matrix and heat wheel shall have an epoxy coating applied to the aluminum matrix. The performance of all wheel prototypes shall be tested in accordance with ASHRAE Standard 84-2020 and **AHRI** Standards 1060-**2018.** The energy wheel shall transfer moisture as well as sensible energy between the two airstreams. The heat wheel shall only transfer sensible energy. The desiccant material shall be a silica gel or

molecular sieve with pore diameter of 3A to minimize the carryover of undesirable gases. Corrugated media shall provide individual flutes to channel the airflow and thus minimize cross contamination and ensure rated performance under all differential pressure conditions. The energy wheel desiccant coating shall provide corrosion resistance against attack from laboratory chemicals, etc. and protection in coastal and marine environments. The heat wheel epoxy coating shall minimize corrosion and latent heat transfer. Wheel rotor media that is 72" diameter or smaller can be either segmented or single piece construction. Wheel rotor media larger than 72" diameter shall be segmented. If segmented, each segment shall consist of media supported by spokes and tensioned in place by the rim. Wheels shall have a flatness of ±1/32". No adhesives or silicone shall be necessary to secure the media in place in order to allow for future replacement in the event of damage to a segment. The structural frame, casing and rotor shall be designed and manufactured so as to allow a maximum rotor deflection of 1/32", as measured at the outer radius, during maximum rated airflow conditions. The media shall have a maximum flame spread of 25 and a maximum smoke developed of 50 when rated in accordance with ASTM E84-**2013a**.

b. Wheel Casing: The wheel frames shall consist of evenly spaced extruded aluminum or corrosion resistant polyurethane coated galvanized steel or epoxy painted heavy duty structural steel spokes, extruded aluminum or corrosion resistant polyurethane coated galvanized steel or epoxy painted heavy duty structural steel outer band, and a rigid center hub. The wheel construction shall allow for wheel alignment. The wheel seals shall be easily adjustable. Seals shall be included to separate the airflows from each other across entire surface of air entering side, air leaving side and outer band (all four planes). Additionally, the entire

circumference of the rotor shall include seal to minimize air bypass. The case construction shall be aluminum or corrosion resistant polyurethane coated galvanized steel or epoxy painted heavy duty structural steel. Bearings shall be outboard-flanged ball bearing with concentric locking collars or tapered roller bearings with double setscrew locking collars. Bearings shall be provided with easily accessible grease fittings for maintenance. Bearings shall provide a minimum L-10 life of 100,000 hours of operation. Sealed bearings are also acceptable.

c. Drive systems shall consist of belt driven heavy-duty inverter rated AC motors. The drive systems shall have a minimum life expectancy of 50,000 hours. The speed control systems shall be variable frequency drive (VFD) capable of operating the rotors from nominally 1/4 RPM to 20 RPM or to the required speed for the type of media used. The OEM controllers for the drive systems shall be integrated with the temperature control system to provide the required energy wheel dry and wet bulb temperatures and heat wheel dry bulb temperatures. The OEM temperature controllers shall monitor entering and leaving temperatures for both airflows. Adjustable set points shall be provided for wheel frost control.

d. Wheels shall be arranged within the heat recovery system so as to minimize differential pressure while keeping the exhaust side negative in comparison to supply side. See drawings for required fan arrangement. The wheels shall have the thickness required to achieve at least a nominal 75% effectivity and shall have a pressure drop not exceeding that as indicated on the Drawings.

e. Wheel Accessories

1. OEM controls and factory set (but field adjustable) energy wheel purge system shall be provided to remove exhaust air

from the depth of the wheel as it revolves from exhaust side to outside air side. The energy wheel purge system shall be provided to eliminate transfer of exhaust air into the supply air.

2. Provide rotation detectors, wheel RPM readouts and wheel stoppage alarm contacts.

3. Provide variable speed drives. The wheels will stop rotation in the economizer mode and modulate in the frost control mode and in the occupied heating modes. Wheels shall run full speed in the occupied cooling mode. Wheels shall be deactivated while the unit is running in a recirculation mode.

f. Maintenance

1. The entire rotor and wheel assemblies shall require only limited maintenance and shall be manufactured not to require more than biannual greasing of the main bearings and inspection of the drive systems.

g. Manufacturers

1. Subject to compliance with requirements, provide enthalpy energy wheel heat exchangers and heat wheel heat exchangers from one of the following manufacturers (or equal). "Equal" indicates that all capacities, dimensions, weights, materials, and performance criteria are equal or better than that described herein.

Thermotech Enterprises TC-Series Thermowheel
Des Champs Thermo-Wheel
AIRotor
Innergy Tech I3

3. Outdoor, Exhaust and Recirculation Dampers

a. Outdoor, exhaust and recirculation dampers shall be of the Class 1 low leakage airfoil blade type with blade edge and side seals that limit leakage to 4 CFM/ft^2 at a pressure differential of 1" WC (Inches of water column) when tested in accordance with AMCA 500D-**2018.** The Class 1 requirements apply to the return recirculation dampers as well as to the outside air intake and exhaust relief dampers per Section C403.5.3.5 of the 2020 NYCECC and Section 6.5.1.1.4 of ASHRAE 90.1-2016. Dampers shall be constructed of extruded aluminum frames (6063T5) of not less than 2.03 mm thickness. Blades shall be of extruded aluminum profiles with blade gaskets of extruded EPDM. Frame seals shall be of extruded Thermoplastic Elastomer (TPE). Gaskets shall be secured in an integral slot within aluminum extrusions. The following dampers shall be provided:

1) Outdoor air damper with electric modulating actuator.

2) Exhaust air damper with electric 2-position actuator (modulation of exhaust flow is accomplished with exhaust fan VFD during Demand Controlled Ventilation). 2-position actuator is used for isolation during the unoccupied modes, warm up mode and pull down mode.

3) Recirculation air damper with electric modulating actuator.

b. Per Section **BC 909 of the 2022 NYC Building Code,** equipment utilized in Smoke Control systems such as automatic dampers shall be suitable for their intended use, suitable for the probable exposure temperatures and shall be as approved by the Commissioner. Smoke control dampers shall be rated and certified by the manufacturer for the probable temperature rise to which the components will be exposed. This temperature rise is _____.

Per Section BC 909.4.6, all portions of active smoke control systems shall be capable of continued operation after detection of the fire event for not less than 20 minutes or 1.5 times the calculated egress time, whichever is more. Per Section BC 909.10.6, Seismic Requirements, smoke control systems covered by Section BC 909 are required to function after an earthquake. Such smoke control systems shall be seismically designed in accordance with Section 1613 and Section 13.2.2 of ASCE 7. The component importance factor, Ip, shall be taken as 1.5 in accordance with ASCE 7, Section 13.1.3. The smoke control system includes all components required for its operation.

4. Supply and Exhaust Fans

a. The supply and exhaust (return) air fans shall be AMCA certified, Class I or II or III as indicated on the Drawing schedule, heavy duty, plenum airfoil type with non-overloading wheels. Flexible duct connections shall be provided to isolate the fan from the cabinet housing as required. Belt driven fan bearing supports shall be constructed of structural steel members to prevent vibration and to rigidly support the fan shaft and bearings. Belt driven fan bearings shall be heavy-duty grease lubricated, self-aligning ball or roller pillow block type. Belt driven fan bearings shall be selected for a basic rating fatigue life (L-10) per AFBMA Standards of 100,000 hours at maximum operating speed for each pressure class and shall be rigidly mounted on welded structural steel members to prevent vibration. Turned, precision ground and polished steel shafts shall be sized so the first critical speed is at least 25% over the maximum operating speed for each pressure class. Direct drive fans shall also be acceptable.

b. Fan performance shall be based on prototypical tests and procedures performed in accordance with AMCA Publication 211-**2016** and Publication

311-**2016** and shall comply with the requirements of the AMCA Certified Ratings Program. Fans shall bear the AMCA seal.

Fan Accessories

1) Lubrication lines shall be extended to a common point on the fan chassis inside the nearest access door.

2) Provide belt guards for belt driven fans to protect personnel from moving drive parts.

3) Provide fan inlet screens for plenum fans.

4) Provide fan outlet enclosures to completely enclose all sides of the fan wheel for plenum fans.

5. Supply and Exhaust Motors and Drives

a. Fan, motor and belt drives shall all be mounted on a spring-isolated chassis (minimum isolation efficiency of 90 - 95%). Belt drives shall have a minimum rated service factor of 1.5. Direct drive fans shall also be acceptable.

b. Motor electrical connections are to be factory pre-wired to the unit control panel. Motors shall be mounted on adjustable bases. 3-phase Open Drip Proof (ODP) type fan motors shall be furnished with NEMA frames and with 1.15 rated service factors. Efficiencies shall be equal to or greater than those specified in the Energy Policy Act of 1992 (EPACT), the 2020 New York City Energy Conservation Code and ASHRAE 90.1-2016 as modified by Appendix CA of the 2020 NYCECC. Fan brake horsepower shall not exceed 90% of motor horsepower. Motors shall be general purpose Inverter Ready motors per NEMA MG1 Part 30.2 designed for a turn down to 20% of full speed. Provide with factory installed grounding rings.

c. Motor starters: Provide variable frequency drive with bypass and circuit breaker for each fan. VFD display shall be mounted in the unit electrical panel.

1) Per Section **BC 909**, equipment utilized in Smoke Control Systems such as fans and motors shall be suitable for their intended use, suitable for the probable exposure temperatures and shall be as approved by the Commissioner. Fans and motors shall be rated and certified by the manufacturer for the probable temperature rise to which the components will be exposed. This temperature rise is _____. Per Section BC 909.4.6, all portions of active smoke control systems shall be capable of continued operation after detection of the fire event for not less than 20 minutes.

2) Per Section BC 909.10.6, Seismic Requirements, smoke control systems covered by Section BC 909 are required to function after an earthquake. Hence, smoke control fans and motors shall be seismically rated. Smoke control systems shall be seismically designed in accordance with Section BC 1613 and ASCE 7, Section 13.2.2. The component importance factor, Ip, shall be taken as 1.5 in accordance with ASCE 7, Section 13.1.3. The smoke control system includes all components required for its operation, including but not limited to fans, ducts, electrical power, switchboards, motor control centers, starters, and controls.

3) Fan Speed on Frequency Drives: At the design flow, motors shall be selected to run continuously at 60 Hz and draw no more than the nameplate horsepower. Any submittal where the submitted operating frequency exceeds 60 Hz shall include an affidavit by the motor manufacturer that the motor can run continuously at the

submitted frequency while providing the design flow. Submittals lacking such affidavit shall be rejected.

4) The supply, return and exhaust fan assemblies, including wheels, shafts, bearings and motors shall be mounted on heavy-duty structural steel support frames bearing on combination spring and neoprene seismic vibration isolators sized and loaded to deflect a minimum of 2”.

6. Filters

a. Both the outside air and return air paths shall be treated with filter banks. The filter banks shall be supplied as integral parts of the packaged air handling unit construction. The outside air path shall be provided with a UL Class 2, 2" thick, pleated type pre-filter with a minimum efficiency of MERV 7 and a final 12” thick cartridge filter with a minimum efficiency of MERV 13. The filters in the outside air path shall be located upstream of the total enthalpy energy recovery wheel. The return air path shall be provided with a UL Class 2, 2" thick, pleated type filter with a minimum efficiency of MERV 13, located upstream of the heat wheel.

b. Access for filter maintenance shall be through a full height double wall hinged service door on the side of the unit. Filters shall be face loaded.

c. Filters shall be selected for a velocity not to exceed 500 fpm. Filters shall comply with UL 900-**2004.**

d. Provide dial type filter pressure gauges to indicate filter pressure drop for each and every outside air filter set and return/exhaust filter.

e. Additionally provide filter differential pressure transducers for each and every

outside air filter set and return/exhaust filter, wired to the OEM controller.

7. Hot water coil (upstream of cooling coil)

a. Hot water coil shall be sized to provide the capacity indicated on the equipment schedule. Tube arrangement shall be staggered. Coils shall have drainable outlet headers. Supply and return connections shall be male pipe thread of the size scheduled. Both supply and return coil connections shall be located at the same end of the coil.

b. Coil shall be internally finned, 0.035" wall copper tubes of the diameter shown on the drawings, mechanically bonded to configured minimum 0.006" aluminum plate fins with a 16 gauge galvanized steel casing. Tube length, number of rows, fins per inch, fin dimensions, airside pressure losses and waterside pressure losses shall be as shown on the Drawing equipment schedules. Coil face velocity shall not exceed 450 feet per minute. Minimum clearance between coil and up/downstream devices shall be 12" free to facilitate cleaning. Coils shall be secured to their respective supports with stainless steel hardware. Coils shall be leak tested at the factory to insure pressure integrity. The coils shall be rated at 250 PSIG water pressure. Coil shall be rated in accordance with ARI standards. Provide factory test reports of leak testing. Vents and drains shall be furnished on all coils.

c. An access door shall be provided on each side of the coil, where feasible, to allow coil removal.

d. All coils over 42" in height shall incorporate a 16 gauge galvanized tube support at the center of the fin height; coils over 96" in fin height shall incorporate additional tube supports. Coils shall be sealed around the perimeter (between the coil flanges and the

unit casing channels) with silicone or polyurethane sealant to eliminate air bypass.

e. OEM or Temperature Controls Contractor (TCC) shall furnish 2-way hot water coil control valve. If OEM furnishes the valve, it may be shipped separately if necessary. Valve shall be installed in the field (if not factory installed by the OEM) by the Mechanical Contractor and wired in the field by the TCC. OEM shall review and approve of the sizing of any TCC furnished valve. Hot water coil control valve shall have an electronic modulating actuator (weather proofed as required) that is spring loaded to open upon power failure.

f. Heating performance of the hot water heating coil shall be based on a 30% propylene glycol water solution.

8. Chilled water coil (downstream of heating coil)

a. Chilled water cooling coil shall be sized to provide cooling/moisture removal of the capacity indicated on the equipment schedule. Tube arrangement shall be staggered. Coils shall have drainable outlet headers. Supply and return connections shall be male pipe thread of the size scheduled. Both supply and return coil connections shall be located at the same end of the coil.

b. Coil shall be of internally finned 5/8″ O.D., 0.035″ wall copper tubes mechanically bonded to configured minimum 0.006″ aluminum plate fins with a stainless steel casing. Tube length, number of rows, fins per inch, fin dimensions, airside pressure losses and waterside pressure losses shall be as shown on the Drawing schedule of equipment. Coil face velocity shall not exceed 450 feet per minute. Minimum clearance between coil and up/downstream devices shall be 12″ free to facilitate cleaning. Coils shall be secured to their respective supports with stainless steel hardware. Coils shall be leak tested at

the factory to insure pressure integrity. The coils shall be rated at 250 PSIG water pressure. Coil shall be rated in accordance with **AHRI** standards. Provide factory test reports of leak testing. Vents and drains shall be furnished on all coils.

c. Coils shall have an integral all seam welded 304 stainless steel sloped drain pan with a minimum depth of 2”. The underneath of the entire drain pan shall be insulated with spray urethane insulation positively preventing any underneath condensate from forming. Entire coil assembly, including safe-offs shall be inside of the recessed drain pan. An access door shall be provided on each side of the coil, where feasible, to allow coil removal.

d. The maximum coil height shall be restricted to 42” to avoid moisture carryover. If necessary, employ multiple face split coils and pans (each coil not exceeding 42” in height) to achieve an effective greater overall coil height. Coils shall be sealed around the perimeter (between the coil flanges and the unit casing channels) with silicone or polyurethane sealant to eliminate air bypass and prevent moisture carryover.

e. OEM or Temperature Controls Contractor (TCC) shall furnish 2-way chilled water coil control valve. If OEM furnishes the valve, it may be shipped separately if necessary. Valve shall be installed in the field (if not factory installed by the OEM) by the Mechanical Contractor and wired in the field by the TCC. OEM shall review and approve of the sizing of any TCC furnished valve. Chilled water coil control valve shall have an electronic modulating actuator (weather proofed as required) that is spring loaded to close upon power failure.

f. Cooling performance of the chilled water coil shall be based on a 30% propylene glycol water solution.

g. Provide OEM limit control freezestat with serpentine element on the inlet of the chilled water cooling coil to stop all fans, prevent cooling coil freeze-ups and prevent cold air from being distributed to the building (to avoid potential building pipe freeze-ups) if the temperature upstream of the chilled water coil is less than or equal to 39˚F (adjustable). Freezestat shall reset automatically unless any trip exceeds 30 seconds as measured by the integral timer. Trips exceeding 30 seconds shall require manual reset.

9. Protection of Components

a. The air handling unit manufacturer shall provide a conditioned sheltered internal environment consistent with the ratings of the equipment for the electrical panels, the environmental DDC controls; disconnect switches and other electrical system components. Provide electrical strip heaters as required to guarantee proper environmental conditions consistent with the ratings of the equipment and components. Ventilation shall be provided as required for all electrical equipment and components using conditioned air stream from the discharge module or exhaust fans. The air handling unit manufacturer shall provide auxiliary heating/cooling as required to guarantee that the required controller operating environmental conditions are provided during all occupied and unoccupied cycles when the equipment cycles intermittently to meet the unoccupied setpoints.

10. Electrical Monitoring/Lighting/Control

a. Provision shall be made by the air handling unit manufacturer for the mounting of current switches on each electrical feeder (Phase A, B, and C) for the unit supply fan, exhaust fan, and main electrical feed. The current switches shall be factory furnished and installed by the OEM. (Current switches are

necessary to obtain fan status when the VFD drive is in the bypass mode).

b. Vapor-proof fluorescent **or LED** lighting shall be installed providing 50-foot candles illumination in the fan and filter sections. All lights shall be powered from a step-down transformer wired to the line side of the main disconnect.

c. An integral electrical control panel shall be provided that has hinged access doors and an approved locking device. All power wiring shall be brought to a common terminal strip and only a single point electrical connection shall be required. Vapor-proof fluorescent **or LED** lighting shall be installed providing 50-foot candles illumination and a fused 5-amp (GFCI) 110-volt receptacle shall be provided. Fused 5-amp receptacle shall be powered independent from the power used for the DDC controller. Provide NEMA Type 4 enclosures constructed for outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, windblown dust, splashing water, and hose-directed water; and that will be undamaged by the external formation of ice on the enclosure.

d. A fused control power transformer shall be furnished. All components shall be fully wired and tested prior to shipment and all major electrical components shall be UL listed. Provide factory test reports. Electrical system shall be ETL listed and labeled, in accordance with UL 1995-**2011** and shall be wired according to NEC. A main disconnect switch shall be factory mounted with rotary handle extending through the unit electrical enclosure. All wiring shall be connected to a numbered terminal strip for easy troubleshooting. Any conduit used shall not be run across or come into contact with the

floor. Enclose all power wiring in liquid tight conduit.

e. Accessories

1. Provide a phase loss monitor. A voltage/phase protection relay shall be provided for each unit. Upon sensing a loss of phase or voltage the unit shall be de-energized.

2. Provide a 20 Amp GFCI convenience receptacle on the outside of the unit powered from a step-down transformer wired to the line side of the main disconnect.

3. A door safety kill switch shall be provided on all fan section access doors. The door safety kill switch shall de-energize the fan motor if the access door is opened. The kill switch shall prevent motor startup if the fan section access door is open.

11. Airflow Measuring Stations - OEM shall provide airflow measuring stations for the total supply flow, (downstream of the recirculation dampers as shown on the control drawings) outside air intake flow and exhaust air flow (located downstream of the total enthalpy wheel). Outside air intake flow and exhaust air flow measurements shall be corrected to account for purge air and shall represent the true outside air provided to the space and exhaust air exhausted from the space. Provide as follows:

a. Provide for low velocity outdoor air applications due to dust and low velocities. Also provide for turbulent locations within placement recommendations of the airflow measuring station manufacturer, where 5 upstream and 1 downstream duct diameters from turbulence causing devices are not available due to physical duct constraints.

b. Each measurement device shall consist of one or more sensor probe assemblies and a single microprocessor-based transmitter. Each sensor probe assembly will contain one or more independently wired sensor housings. Multiple independent sensors shall produce measurements equally weighted and averaged by the transmitter prior to output. Pitot tubes and arrays are not acceptable. Vortex shedding airflow meters are not acceptable.

c. All airflow/temperature measurement sensor probe assemblies shall comply with the following:

1) Each sensor housing shall utilize hermetically sealed, bead-in-glass type thermistors or RTD probes used to determine airflow rate and ambient temperature.

2) Each independent sensor-housing shall be calibrated at a minimum of sixteen (16) airflow rates and have an accuracy of ±2% of reading over the entire operating airflow range. Each sensor assembly shall be factory calibrated to standards that are traceable to the National Institute of Standards and Technology (NIST).

3) Devices whose accuracy is the combined accuracy of the transmitter and sensor probes must demonstrate that the total accuracy meets the performance requirements of this specification throughout the measurement range.

4) The operating temperature range for the sensor probe assembly shall be -20°F to 160°F. The operating humidity range for the sensor probe assembly shall be 0-99% RH (non-condensing).

5) Each temperature sensor shall be calibrated at a minimum of 3 temperatures and have an accuracy of ±0.15°F over the entire operating temperature range. Each

temperature sensor shall be factory calibrated to standards that are traceable to the National Institute of Standards and Technology (NIST).

6) Each sensor probe assembly shall have an integral, U.L. listed, plenum rated cable and terminal plug for connection to a remotely mounted transmitter. All interconnecting pins shall be gold plated.

7) Each sensor assembly shall not require matching to the transmitter in the field.

8) A single manufacturer shall provide both the airflow/temperature measuring probe(s) and transmitter at a given measurement location.

9) The number of sensor housings provided for each location shall be as follows:

AREA (SQ.FT.)	Sensors
<=1	2
>1 TO <4	4
4 TO <8	6
8 TO <12	8
12 TO <16	12
>16	16

10) Probe assembly mounting brackets shall be constructed of stainless steel. Probe assemblies shall be mounted using one of the following options:

a) Insertion mounted (through the side or top of the duct)

b) Internally mounted (inside the duct or plenum)

c) Standoff mounted (inside the plenum, immediately upstream of an intake damper or downstream of a filter or coil bank.) with a probe extending into the outside air flow stream.

11) Range

The operating airflow range shall be 0 to 5,000 FPM unless otherwise indicated on the plans.

12) The transmitters shall have an LCD display capable of displaying airflow and temperature. Airflow shall be field configurable to be displayed as a velocity or a volumetric rate.

13) The transmitter shall be capable of displaying the individual airflow and temperature readings of each sensor on the LCD display.

14) The transmitter shall operate on 24 VAC or 24 VDC. The transmitter shall not require an isolated power source.

15) The operating temperature range for the transmitter shall be -20°F to 120°F. The transmitter shall be protected from weather and water.

16) The transmitter shall be capable of communicating with the OEM host controls using one of the following interface options: Linear analog output signal: Field selectable, fuse protected and isolated, 0-10 VDC and 4-20mA (4-wire).

Airflow measuring station transmitters in both the total supply and exhaust air streams shall send signals to the OEM controller which in turn shall send

signals to the supply fan VFD and exhaust fan VFD to maintain the flow setpoint to within +/- 5% of flow setpoint.

17) The airflow/temperature measuring device shall be UL listed as an entire assembly.

18) The airflow measuring station manufacturer's authorized representative shall review and approve placement and operating airflow rates for each measurement location indicated on the air handling unit OEM shop drawings.

19) A written report shall be submitted to the Authority's Architect/ Engineer-of-Record/ Commissioning Authority if any measurement locations do not meet the airflow measuring station manufacturer's placement requirements.

20) Acceptable Manufacturers (or approved equal):

Ebtron Series GP1 Duct and Plenum Probe
Dwyer PF Series Precision Flow Air Velocity Transmitter

12. Digital Controls (furnished, installed, wired, configured, programmed, and commissioned by the Original Equipment Manufacturer (OEM)).

a. An electronic programmable microprocessor-based OEM BACnet controller shall be factory furnished, installed, wired, configured, programmed and commissioned to control the air handling unit. Refer to Section {15970 15971 15975} for specifications of the temperature control system instrumentation. Field commissioning of the units shall be conducted in the presence of the Commissioning Authority and performed by representatives of the manufacturing company, mechanical contractor (MC), and the Temperature Controls Contractor (TCC). Refer to Section 15973 for additional information.

b. All wiring of sensors and controlled devices shall be by the OEM with the exception of the outside air temperature, outside air relative humidity and demand controlled ventilation outside air carbon dioxide sensors and space temperature, space relative humidity and demand controlled ventilation space carbon dioxide sensors which shall be furnished by OEM in accordance with Section {15970 15971 15975}, installed and wired by the Temperature Controls Contractor (TCC) to the air handling unit controller. In addition, the TCC shall wire the hot water and chilled water coil control valves since they will be shipped separately. OEM (or OEM's authorized field service representative) shall be responsible for field commissioning of the controls i.e. demonstrating their proper sequence of operation to the Commissioning Authority in accordance with the sequence of operation as defined in Section 15985, Sequence of Operations.

c. The temperature control system shall be a BACnet compatible microprocessor-based controller fully capable of stand-alone direct digital operation utilizing its own processor, non-volatile memory, input/output, A to D conversion, clock/calendar and voltage transient and lightning protection devices. All non-volatile memory shall have a battery backup of at least seventy-two (72) hours. The DDC controls shall not be dependent on communications with an on-site network system and shall be equipped with on-board diagnostics, indicating that all hardware, software and interconnecting wiring are in proper operating condition. Each OEM controller shall have built-in networking capability so that multiple controllers may be easily networked together via the BACnet network using the open BACnet protocol. Each OEM controller shall perform its own self-test diagnostics, that shall be automatically relayed over the BACnet MS/TP or BACnet IP network to report any malfunctions or alarm

conditions that exceed desired, custom-definable parameters.

d. A large 4-line by 40-character backlit LCD display shall be provided for easy reading even in poor lighting conditions. Display shall include a numeric keypad, directional keys, four programmable function keys and an alarm indicator light with audible warning. Temperature and humidity set points shall be easily input and adjusted by the operator. All programming tool-sets, interface cables and controller configuration tools used to configure the system operation and generate the control database shall be provided with the unit control system. OEM shall provide a licensed copy of all tool sets to the Authority for use with this system.

e. All required air handling unit temperature sensors and relative humidity sensors shall be provided by the OEM as indicated in the control sequence. Outside humidity sensor, space humidity sensor, outside temperature sensor and space temperature sensor, demand controlled ventilation outside carbon dioxide sensor and demand controlled ventilation space carbon dioxide sensor shall be furnished by OEM in accordance with Section {15970 15971 15975} and field mounted and wired by the Temperature Controls Contractor (TCC).

f. Unit controls shall be compatible with the BACnet network by virtue of the controller being BACnet compliant. Unit manufacturer shall provide complete controls sequence with points list in the submittal package. Field mounted unit controls shall be unacceptable as they prevent full factory testing and may substantially compromise the unit operation.

g. All sensors and transducers shall report to the OEM controllers and the associated defined network variables shall be passed directly on to or from the BACnet network by BACnet OEM controller. Any required points shown on the

Control Diagrams/Points Lists that cannot be transmitted by OEM's controller, shall be provided by TCC field provided sensors and transducers which shall be connected to a BACnet controller that has spare I/O in order to get the required points on to the BACnet network. OEM shall coordinate with TCC by highlighting any points that cannot be provided through OEM's controllers and which must be provided by the TCC.

h. In general, the OEM controls shall modulate the heating coil hot water valve, cooling coil chilled water valve; outside air, exhaust air, and recirculation dampers; total enthalpy wheel and heat wheel VFD drives; and supply/exhaust fan VFDs based on occupied/unoccupied scheduling.

i. Each OEM controller shall contain an asynchronous serial port for local access via a laptop computer or Palmtop type computer. Communication through this port shall allow for local access to that particular controller. Access shall be password protected with various levels of definable operator access. Once a proper password has been entered, the user (depending upon his or her password level) shall be able to view all status and alarm conditions as well as make changes to the operating parameters, such as setpoints and schedules.

j. The OEM and TCC shall provide all system passwords for system configuration and programming to the Department of Education and a full system database backup provided to the Department of Education on CD media. Two copies of all system control data shall be provided; one maintained on-site by the local school custodian and an additional copy for off-site storage by the Department of Education.

k. Input-Output Processing:

See Sections {15970 15971 15975} and 15985 for Input and Output information. Input-Output Processing shall be as follows:

1) Digital outputs from OEM controller: Outputs shall be 24VAC or VDC maximum, 3-amp maximum current. Each output shall be configurable as normally open or normally closed, and either dry contact or bussed.

2) Universal inputs to OEM controller: All input points shall be universal, and shall be individually jumper selected to serve as either a digital input (DI) or analog input (AI). The acceptable input types are: Thermister - 10K Ohm at 77°F, 0-5VDC - 10K Ohm maximum source impedance, 0-20mA — 24 VDC loop power 250 Ohm input impedance, or Dry Contact - 0.5mA maximum current.

3) Analog output electronics from OEM controller shall be: voltage mode, 0-10 VDC; current mode, 4-20mA.

l. Operating and Safety Controls (Provided by OEM except as noted otherwise): Provide factory wired control panel containing but not limited to the following:

1) Contactors for each fan motor.

2) Supply fan static over pressurization switch and exhaust fan under pressurization switch

3) Overload protection for each motor.

4) Supply air and return air reset control

5) Emergency stop control input

6) Occupied/Unoccupied switching

7) Timed override activation.

8) Post-Fire Smoke Purge control and ventilation override sequence (By Fire Alarm Contractor)

m. OEM return air temperature sensor and return air relative humidity sensor shall be supplied with calibration certificates indicating prior factory calibration to reference standards that are traceable to the National Institute of Standards and Technology (NIST).

E. Points List and Sequence of Operations

1. See the bid document Control Details for the required points list.

2. See Section 15985 for the Sequence of Operation.

F. Fire Alarm Interface Requirements

The packaged air handling unit shall have UL 268A-**2009** (per Section MC 606.1) manual reset listed type smoke detectors factory mounted and wired in the supply and return openings to the unit as required by the NYC Construction Codes and Electrical Code (i.e. when the flow is greater than 2,000 cfm). Upon detection of smoke, a signal shall be sent to the building fire alarm system and the fans shall stop as commanded by the fire alarm control panel. Installing Mechanical Contractor shall coordinate this interlock with the Division 16 Fire Alarm Contractor. Division 16 Fire Alarm Contractor shall furnish fire alarm system compliant photoelectric smoke detectors and individual fire alarm system compliant fan and HVAC controls shutdown relays to the Mechanical Contractor who shall in turn furnish them to the air handling unit manufacturer for factory installation. Per NFPA 72-**2016,** fire alarm system compliant fan shutdown relays shall be mounted within 3' of the motor starters/VFDs and wired into the safety circuits to stop the fans in all modes. Fire alarm relays pre-wired to the HVAC unit inputs shall allow unit to index non-fan devices to System-Off Mode. Detectors shall be rated from -4°F to 158°F and 100 fpm to 4,000 fpm and shall be furnished with a weatherized housing to prevent condensation from humidity. As per

NFPA 72, it is acceptable to have a supervised auxiliary fan shut down relay within 3′-0″ of interposing UL 864-9th edition listed relays on the same ladder rung of the ladder logic that connects to the starters to shut them down directly. Use of cascading series of relays beyond the permitted interposing relay(s) to shut down the fans is not code compliant.

G. Acoustical Features for the Basis of Design:

Perforated lining shall be provided in all fan sections for additional sound attenuation. In addition, the following shall be provided:

1. Bases shall be designed to accept point load vibration isolators. Internal isolation of rotating components [shall] [shall not] be provided.

2. Acoustically lined double walled outdoor air intake and exhaust air hoods with perforated liners sandwiching the 2 lb/ft^3 density, acoustic lining of minimum 2″ thick polyurethane foam shall be provided.

3. Silencer Section: Acoustical silencer sections (in addition to the perforated lining of all of the fan sections) shall be incorporated into the air handling unit. The required silencer length and acoustical performance shall be as shown on the Schedule of Equipment.

   Acceptable suppliers:

   Industrial Acoustics Corporation
   AeroAcoustics
   Vibroacoustics
   Ruskin

4. If the contractor elects to submit units other than the Basis of Design for shop drawing review, units shall be provided with internal or external vibration isolators; and/or acoustically lined double walled outdoor air intake and exhaust air hoods with perforated liners sandwiching the 2 lb/ft^3 density acoustic lining of minimum 2″ thick polyurethane foam; and/or acoustical silencer section supplied by Industrial Acoustics

Corporation or AeroAcoustics or Vibroacoustics or Ruskin as required to meet the below maximum allowable sound power levels.

The maximum radiated sound power levels shall be as follows:

Center Frequency (Hz)	63	125	250	500	1000	2000	4000	8000
Band Designation	1	2	3	4	5	6	7	8
dB								

The maximum supply discharge sound power levels shall be as follows:

Center Frequency (Hz)	63	125	250	500	1000	2000	4000	8000
Band Designation	1	2	3	4	5	6	7	8
dB								

The maximum return inlet sound power levels shall be as follows:

Center Frequency (Hz)	63	125	250	500	1000	2000	4000	8000
Band Designation	1	2	3	4	5	6	7	8
dB								

H. Smoke Dampers for Units of 15,000 CFM and Greater Capacity: Not used.

2.02 MANUFACTURERS

A. Subject to compliance with requirements, provide packaged air handling units from one of the following manufacturers (or equal). “Equal” indicates that all capacities, dimensions, weights, materials, and performance criteria are equal or better than basis of design. All costs incurred by design deviation shall be the responsibility of the mechanical contractor, including operating costs.

Munters DesChamps
Annexair
Xetex
Seasons 4
Coolbreeze
Mafna

PART 3 – EXECUTION

3.01 SUPPLEMENTAL INSTALLATION

A. Packaged Air Handling Units

1. Electrical Wiring: The Division 16 Electrical contractor shall provide power for the units. Furnish copy of manufacturer's wiring diagram submittal to Electrical.

   a. All wiring in the packaged air handling unit sections shall be in standard (in accordance with the NYC Electrical Code) or flex type conduit. Wiring shall be color-coded and shall be numbered at termination points. Control circuit wiring shall be 16-gauge minimum type TFF. Power wiring shall be minimum 12-gauge type THHN wire. Wiring diagrams shall be laminated to inside of control panel door. The air handling unit manufacturer shall provide control panel(s) that shall contain all necessary components for full control of the unit. These control panels shall be provided with 115 volts and 24 volts step down transformers.

   b. The air handling unit manufacturer shall furnish, controllers with sufficient input and output (I/O) capability for system operation, monitoring and alarms. Interface relays shall be provided. Shop drawings shall illustrate complete unit interface wiring diagrams for the specific unit and testing procedures. Shop drawings shall indicate the Sequence of Operations.

   c. The air handling unit manufacturer shall provide all safety controls to meet code requirements, a 24-volt control circuit, and

mounting and wiring of all unit controls and devices. Air handling unit manufacturer shall furnish terminal strip(s) to interface the OEM DDC Controller to the field mounted devices to be field wired by the Temperature Control Contractor (TCC).

2. Ductwork: Refer to Section 15891: Metal Ductwork. Connect supply and return/exhaust ducts to unit with flexible duct connections. Provide transitions to exactly match unit duct connection size.

3. Drain Piping: Connect unit drain to splash block. Provide trap at drain pan; construct at least 1" deeper than fan pressure in inches of water.

3.02 FIELD QUALITY CONTROL

A. Start-up packaged air handling units in accordance with manufacturer's start-up instructions and in the presence of a manufacturer's representative. The OEM and Temperature Control Contractor shall test the OEM DDC system and communication cards and demonstrate compliance with requirements of the Specifications and Drawings to the satisfaction of the Commissioning Authority. Replace all damaged and malfunctioning controls and equipment. The Contractor shall perform necessary Interdisciplinary Tests and Functional Performance Tests according to the manufacturer’s procedures and as described in Section {15971}.

3.03 SUPPLEMENTAL TRAINING

A. Provide services of manufacturer's technical representative and the Temperature Control Contractor for 20 hours to instruct the Custodian (or building manager) and DOE Maintenance Representative in the operation and maintenance of packaged air handling units.

B. Schedule training with Authority and provide at least two days notice of training date to the Custodian and DoE Maintenance Representative.

C. The documentation for each controller shall include an accurate graphic flow diagram of the software program for future owner reference.

3.04 INTERDISCIPLINARY TESTS AND FUNCTIONAL PERFORMANCE TESTS

A. Interdisciplinary Pre-Start-Up and Start-Up Tests

The Contractor shall conduct interdisciplinary pre-start up and start up tests as per the manufacturer's start up procedures. Contractor shall submit signed start up affidavit signed by the factory authorized service representative indicating that all of the manufacturer's pre-start up and start up procedures have been successfully completed.

B. Functional Performance Tests

Contractor shall also submit signed functional performance testing affidavit signed by the factory authorized service representative indicating that all of the manufacturer's functional performance tests have been successfully completed. Refer to the functional performance tests as defined in Section {15970 15971 15975}.

3.05 COMMISSIONING OF AIR HANDLING HVAC UNITS

A. HVAC Contractor shall comply with the Commissioning Requirements of Contract Specification Section S01660 for air handling HVAC units.

END OF SECTION

BB:RF
11/30/22

LIST OF SUBMITTALS

SUBMITTAL	DATE SUBMITTED	DATE APPROVED
Product Data:	_____	_____
1. Manufacturers Product Data (including motor manufacturer affidavit if greater than 60Hz operation)		
2. Installation Instructions		
3. Performance Charts		
Shop Drawings:	_____	_____
1. Assembly-type		
2. Mounting Details		
3. Location		
4. Connection Details		
5. Wiring Diagrams:	_____	_____
a. Power Wiring		
b. Control Wiring		
c. BACnet Control Interface and Control Wiring with Sequence of Operation		
Coordination Drawings:	_____	_____
Certifications:	_____	_____
1. Fan-Sound Power Ratings		
2. Contractor's Start-Up and Demonstration Affidavit		
3. Seismic Certificate of Operability		
Maintenance Data:	_____	_____
1. Maintenance Manual		
Maintenance Materials:	_____	_____
1. Spare Filters Set		
2. Spare Belt Set	_____	_____
Preventative Maintenance Plan and Schedule	_____	_____

Manufacturer's Guarantee/ Warranty	_____	_____
Manufacturer's Factory Test Results	_____	_____
Toolset (software) for OEM Controller	_____	_____

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